Python Assignment
Topic-5: Looping (For, While)

Q1. Introduction to for and while loops.

Ans.

Loops are used to repeat a block of code multiple times. The 'for' loop is used when we know how many times we want to repeat the code. It is commonly used with lists, strings, or ranges. The 'while' loop is used when we do not know the number of repetitions in advance. It runs until a condition becomes false. Loops help in reducing code repetition and make programs shorter and efficient. They are very useful in solving problems that require repeated actions.

Example:

```
for i in range(5):
   print(i)

x = 1
while x <= 5:
   print(x)
   x += 1
```

Q2. How loops work in Python.

Ans.

Loops work by executing a block of code again and again based on a condition. In a 'for' loop, Python takes each value from a sequence and runs the code for each value. In a 'while' loop, Python checks the condition before each iteration. If the condition is true, the loop continues. If it becomes false, the loop stops. Loops make tasks like printing numbers, processing data, and repeating actions easy. Proper use of loops makes programs more efficient and readable.

Example:

```
numbers = [1,2,3]
for n in numbers:
   print(n)
```

Q3. Using loops with collections (lists, tuples, etc.).

Ans.

Loops can be used with collections like lists, tuples, and sets to access each element one by one. This helps in performing operations on multiple values easily. Using loops with collections saves time and reduces code length. It is commonly used in data processing and handling multiple values. We can apply conditions inside loops to filter data. This makes loops very powerful in Python programming. It is a basic and important concept for beginners.

Example:

```
fruits = ["apple", "banana", "mango"]
for f in fruits:
    print(f)
```